Architectural Acoustics

Acoustics - Physics dealing with the generation, propagation and applications of sound. Acoustics is the science of sound, which deals with the properties of sound waves.

Architecture of Acoustics - deals with the design and construction of music halls and sound recording rooms to provide the best audible sound to the audience.

- Sound is a vibration in an elastic medium with a definite frequency and intensity that can be heard by the human ear.

CLASSIFICATION OF SOUND:-

1. Infra sound (< 20 Hz)
2. Audible sound (20 Hz – 20K Hz)
3. Ultrasound (>20K Hz)

Classification of audible sound:-

(a) Musical Sound
(b) Noise

(a) Musical Sound: - The sound that produces a pleasing effect on the human ear is called musical sound. Example: -sitar, violin, flute, piano, etc.

Properties of musical sound:

1. Regular waveforms.
2. Definite periodicity.
3. No sudden change in amplitude.

(b) Noise: - The sound that produces an unpleasant effect on the human ear is called noise. Example: - road traffic, crackers, etc.

Properties of Noise: -

1. Irregular shape of waveforms
2. No definite periodicity.
3. Sudden change in amplitude.

CHARACTERISTICS OF MUSICAL SOUND:-

1. **Pitch:** - related to the frequency of sound
2. **Loudness:** - related to the intensity of sound
3. **Timbre:** - related to the quality of sound

Some Definitions:

Pitch: - pitch is the related frequency of sound. Pitch helps distinguish between a high-frequency and a low-frequency sound of the same intensity produced by the same musical instrument.

- The sound produced by mosquitoes and bees is of high pitch due to high frequency.
- Thus, the greater the frequency of a sound, the higher the pitch and vice versa.

Loudness: - a loudness characteristic which is common to all sound, whether classified as musical sound or noise.

- Loudness is a degree of sensation produced in the ear. Thus, loudness varies from one listener to another.
- Loudness depends on the intensity of sound and relies on the sensitivity of the ear. Relation between loudness and intensity:-

  L=K log I, where K is a constant.
- Loudness is directly proportional to the logarithm of intensity. The above relation is known as the **Weber-Fechner law.**
- On taking the differentiation of the above relation, we get, which is termed the sensitivity of the ear,

$$dL/dI = K/I$$

  Therefore, sensitivity decreases with an increase in intensity.
- Loudness is a physiological quantity.

Timber: The quality of sound that enables us to distinguish between two sounds having the same loudness and pitch. It depends on the presence of overtones.

It helps in distinguishing between musical notes emitted by different musical instruments and the voices of different persons, even though the sounds have the same pitch and loudness.

<u>Intensity(I)</u>:- Intensity of sound waves at a point is defined as the amount of the sound energy Q flowing per unit area in unit time.

$$I=Q/At$$

Where A= area, t = time and Q = amount of the sound energy If A=1 $m^2$and t= 1 sec, then I=Q

Intensity is a physical quantity that depends on factors like amplitude (a), frequency (f), and velocity of sound, together with the density of the medium ρ.

$$I= 2\Pi^2 f^2 a^2\ \rho v$$

The unit of intensity is Watt/m^2.
The minimum Intensity that a human ear can sense is called the threshold intensity.
Its value is 10^{-12} watt/m^2.
This minimum intensity is also known as zero or standard Intensity.

Intensity level (Relative intensity):-

The Intensity level or Relative intensity of a sound is defined as the logarithmic ratio of the intensity I of a sound to the standard intensity I_0.

$$I_L = K \log (I/I_0)$$

Let I and I_0 represent two sounds of a particular frequency, and L_1 and L_0 represent loudness. According to the **Weber-Fechner law,**

$$L_1= K \log I$$
$$L_0= K \log I_0$$

Therefore, the intensity level or Relative intensity is

$I_L= L_1 - L_0 = K \log I – K \log I_0$
$= K (\log I – \log I_0)$
$I_L = K \log (I/I_0)$ bel

If K=1, then I_L is expressed in a unit called bel.

I=10 I_0, 1 bel
I=100 I_0, 2 bel
I=1000 I_0, 3 bel

Bel is a large unit, hence another unit known as decibel (dB)

1dB= $\frac{1}{\ }$ bel

10

$$I_L = 10 \log (I/I_0)\ dB$$

Sound Absorption: The property of a surface by which the incident sound energy is converted into other forms of energy is called absorption.

SOUND ABSORPTION COEFFICIENT ‘a’

Definition 1: The sound absorption coefficient ‘a’ of a material is defined as the ratio of sound energy absorbed by it to the total sound energy incident on it.

Absorption coefficient a $=\frac{\textit{Sound energy absorbed by the surface}}{\textit{total sound energy incident on it}}$

Definition 2: It is defined as the reciprocal of the area of sound-absorbing material that absorbs the same amount of sound energy as that of 1 m^2 of an open window.

The unit of absorption coefficient is Sabine and O.W.U.(Open Window Unit)

Reverberation

Reverberation is the property of sound that determines how long one can hear sound even after the speaker has stopped speaking/producing sound.

Or

The persistence or prolongation of sound in a hall, even though the source of sound is cut off, is called reverberation.

Reverberation Time

The time interval during which sound can persist after the speaker stops sound.

Sabine’s Formula for Reverberation Time

The time taken by the sound intensity to fall to one millionth of its original intensity after the source stopped emitting sound.

$$T = \frac{0.167V}{\sum a.s}$$

Here, V = volume of hall, a = absorption coefficient, and S = surface area.

It is directly proportional to the volume of the hall and inversely proportional to the total absorption made by the hall.

Average absorption coefficient $\boldsymbol{a = \frac{a_1 s_1 + a_1 s_1 +--}{s_1 + s_2 + --} = \frac{\sum a_i s_i}{\sum s_i}}$

FACTORS AFFECTING ACOUSTICS OF BUILDINGS AND THEIR REMEDIES:-

The various factors affecting the acoustics of buildings, such as reverberation time, loudness, focusing, echo, echelon effect, resonance, and noise, with their remedies, are explained in brief in this section.

1. **Reverberation Time :**
   - Reverberation is the persistence or prolongation of sound in a hall even after the source has stopped emitting sound.
   - The reverberation time is the time taken by the sound to fall below the minimum audibility level.
   - To have a good acoustic effect, the reverberation time has to be maintained at an optimum value.
   - The reason is, if the reverberation time is too small, the loudness becomes inadequate. As a result, the sound may not reach the listener. Thus, this gives the hall a dead effect.
   - On the other hand, if the reverberation time is too long, it will lead to more confusion due to the mixing of different syllables. This makes the sound unintelligible.
   - Thus, reverberation time should neither be too large nor too small.
   - Hence, to maintain a good acoustic effect, the reverberation time should be maintained at an optimum value.

Remedies: The reverberation time can be maintained at an optimum value by adopting the following methods:

1. By providing windows and openings.
2. By having a full capacity audience in the hall or room, the Architectural Acoustics
3. By using heavy curtains with folds.
4. By covering the floor with carpets,
5. By decorating the walls with beautiful pictures, maps, etc.
6. By covering the ceiling and walls with good sound-absorbing materials like felt, fiberboard, false roofing, etc.

The reverberation time depends on the size of the hall and the quality of sound. Thus, the reverberation time can be controlled either by inserting or removing sound-absorbing material in a hall or room.

2. **Loudness**

The uniform distribution of loudness in a hall or a room is an important factor for satisfactory hearing. Sometimes, the loudness may be reduced due to an excess of sound-absorbing materials used inside a hall or room.

Remedies: If the loudness of the sound is not adequate, the loudness can be increased by adopting the following methods.

1. By using suitable absorbents at places where noise is high. As a result, the distribution of loudness may become uniform.
2. By constructing low ceilings for the reflection of sound towards the listener.
3. By using large soundboards behind the speaker and facing the audience.
4. By using a public address system like loudspeakers.

3. **Focusing and Interference Effects**

- The presence of any concave surface or any other curved surface in the hall or room may cause the sound to be concentrated at this focus region.
- As a result, the sound may not be heard at all in other regions. These regions are referred to as dead space. Hence, such surfaces must be avoided.
- In addition to focusing, there should not be interference of direct and reflected waves. This is because constructive interference may produce a sound of maximum intensity in some places, and destructive interference may produce a sound of minimum intensity in other places.
- Thus, there will be an uneven distribution of sound intensity.

Remedy

Curved surfaces can be avoided. If curved surfaces are present, they should be covered with suitable sound-absorbing materials.

4. **Echo:**

- An echo is heard due to the reflection of sound from a distant sound-reflecting object.
- If the time interval between the direct sound and reflected sound is less than $1/15^{th}$ of a second, the reflected sound helps increase the loudness. But those sounds arriving later than this cause confusion.

Remedy

An echo can be avoided by covering long-distance walls and high ceilings with suitable sound. Absorbing material. This prevents the reflection of sound.

5. **Echelon Effect**

It refers to the generation of a new, separate sound due to multiple echoes. A set of railings or any regular reflecting surface is said to produce the echelon effect. This echelon effect affects the quality of the original sound.

Remedy

The remedy to avoid the echelon effect is to cover such surfaces with sound-absorbing material.

6. **Resonance**

Resonance occurs due to the matching of frequency. If the window panels and sections of wooden portions have not been tightly fitted, they may start vibrating, thereby creating an extra sound in addition to the sound produced in the hall or room.

Remedy

The resonance may be avoided by fixing the window panels properly. Any other vibrating object that may produce resonance can be placed over a suitable sound-absorbing material.

7. **Noise**

The unwanted sound is called a noise. The hall or room should be properly insulated from external and internal noises. In general, there are three types of noise:

1. **Air-borne noise,**
2. **Structure-borne noise,**
3. **Inside noise.**

1. Air-borne Noise

Extraneous noises that are coming from outside through open windows, doors and ventilators are known as air-borne noise. The air-borne noise can be avoided by following the remedies mentioned below.

1. The hall or room can be made air-conditioned.
2. By using doors and windows with separate frames with proper sound-insulating material between them.

2. Structure-borne Noise

The noise that is conveyed through the structure of the building is called structure-borne noise. The structural vibration may occur due to street traffic, the operation of heavy machines, etc.

Remedies

1. This noise can be eliminated by using double walls with an air space between them.
2. By using anti-vibration mounts, this type of noise can be reduced.
3. By covering the floor and walls with proper sound-absorbing material, this noise can be eliminated.

Inside Noise

- The noises that are produced inside the hall or room are called inside noise. The inside noise may be produced due to machinery like air conditioners, refrigerators, generators, fans, typewriters, etc.
- To avoid these inside noises, the following remedies can be adopted.

Remedies

1. The sound-producing machineries can be placed over sound-absorbing materials like carpet, pads, wood, felt, etc.
2. By using curtains of sound-absorbing materials.
3. By covering the floor, wall and ceiling with sound-absorbing materials.

3.9.1 Conditions for Good Acoustics

A hall or an auditorium is said to be acoustically good if it satisfies the following conditions:

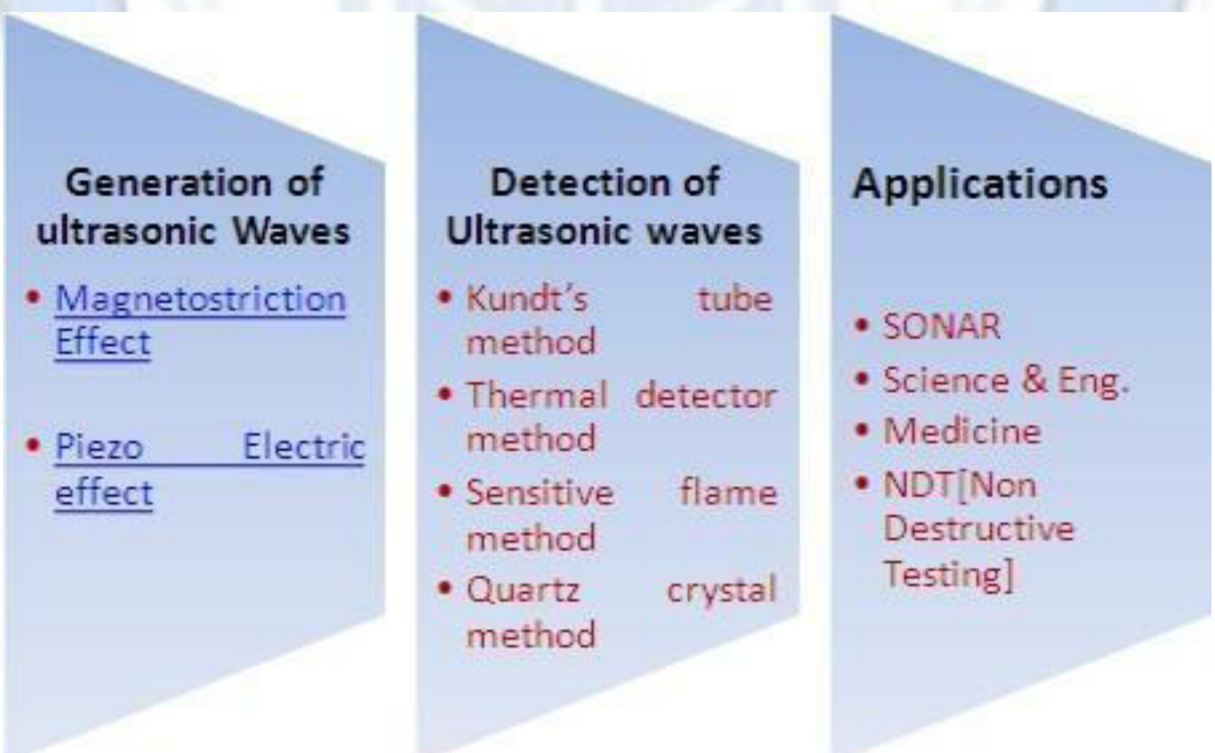
1. The quality of the sound should be uniform throughout the entire hall or auditorium.
2. There should not be any overlapping of sounds.
3. The loudness of the sound should be uniform throughout the hall or auditorium. To achieve this, a public address system can be used in big halls.
4. The presence or absence of an audience should not affect the quality of sound.
5. The resonance effect should be avoided.
6. The hall should have a proper reverberation time.
7. The external noises should not disturb the proceedings inside the hall or auditorium.
8. There should not be any echelon effect.

Ultrasonics

Ultrasonic Waves: The sound wave with a frequency greater than 20 KHz is called an Ultrasonic wave.

Properties of Ultrasound:

- Frequency is greater than 20KHz
- It can propagate long distances because of its high frequency
- It can travel with constant speed in a homogeneous medium. It cannot travel in a vacuum.
- As the wavelength of ultrasonic waves is shorter, their frequency is higher, and their penetration power is higher.
- The speed of an ultrasonic wave is larger in a denser medium.
- Since it is a longitudinal wave, it is not possible to polarize them.
- Travel through liquid: longitudinal nature
  Travel through a solid: longitudinal as well as transverse nature.
- Humans cannot hear it, but some animals can hear it (dogs, bats, and dolphins)

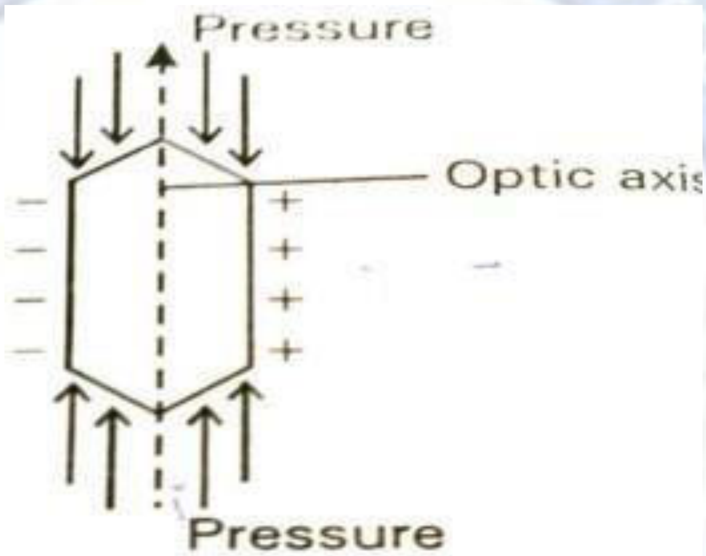


Determination of Ultrasonic waves

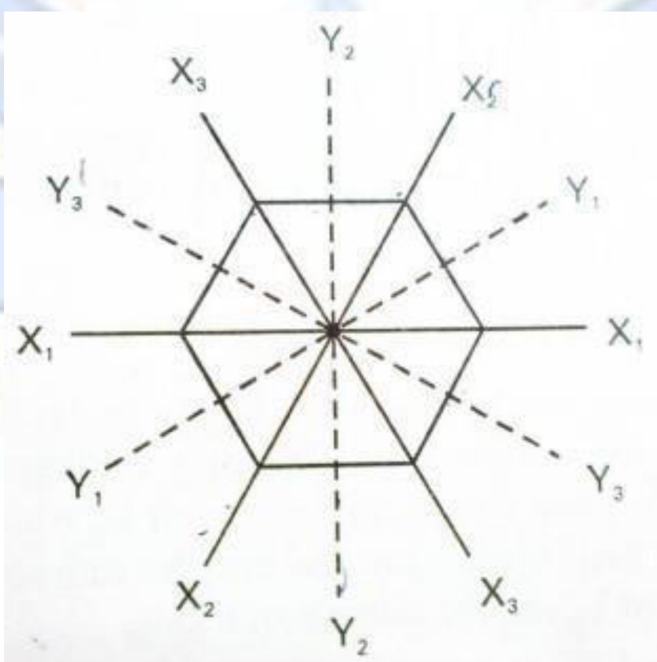
Piezoelectric Method:

Piezoelectric effect:

When pressure is applied to one pair of opposite faces of crystals like Quartz, tourmaline, Rochelle salt, etc, cut with their faces perpendicular to their optic axis, equal and opposite charges appear across their other faces, as shown in the figure. This is known as the piezoelectric effect.



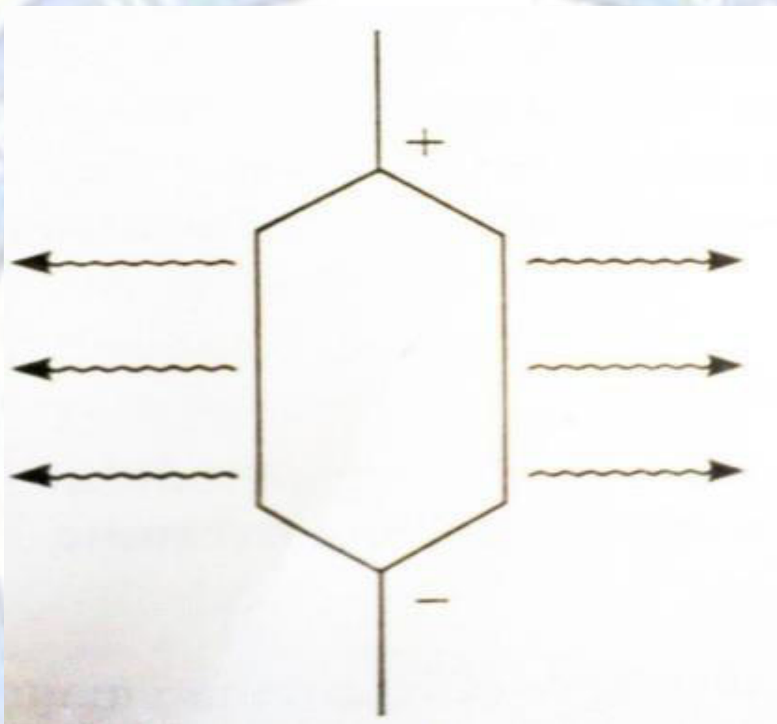
The cross-section of a natural quartz crystal is hexagonal. The lines joining the opposite corners of the hexagon are called the electric axes (X_1, X_2, X_3) and the lines joining the midpoint of the opposite faces of the hexagon are called the mechanical axes (Y_1, Y_2, Y_3)



Inverse piezoelectric effect:

If an alternating voltage is applied to one pair of opposite faces of the crystal, alternatively, mechanical contractions and expansions are produced in the crystal, and the crystal starts vibrating. This effect is known as the **inverse piezoelectric effect** or **electrostriction effect**.

When the frequency of the applied alternating voltage is equal to the vibrating frequency of the crystal, then the crystal will produce ultrasonic waves.



Piezoelectric oscillator method for Inverse piezoelectric effect

Construction:

Quartz crystal Q is placed between two metal plates A and B, connected to the coil L3. The coils L1, L2 and L3 are inductively coupled to the oscillatory circuit of a transistor. Coil L2 is connected to the collector circuit, while coil L1 with a variable capacitor C1 forming the tank circuit. The high-tension battery is connected between the free end of L2 and the emitter of the transistor.

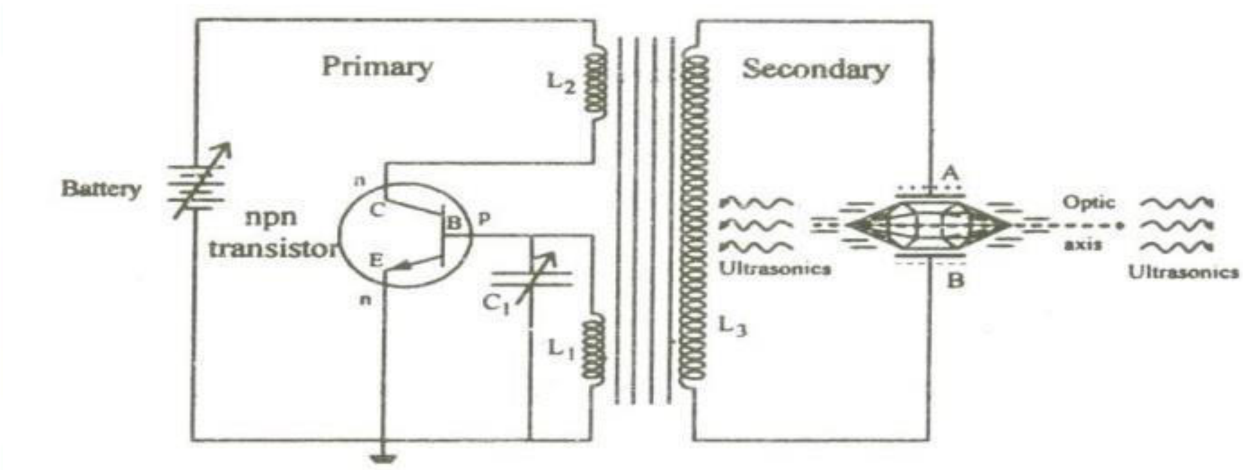


Figure Piezo-Electric Oscillators

Working:

When the battery is switched on, the oscillator produces a high-frequency alternating voltage given by,

- The frequency of oscillation can be controlled by the variable capacitor C_1.

$$f = \frac{1}{2\pi\sqrt{L_1 C_1}}$$

- Due to the transformer action, an emf is induced in the secondary coil L_3. This emf impressed on the plates A and B will excite the quartz crystal into vibrations.
- By adjusting the variable capacitor C_1, frequency can be achieved in resonant conditions crystal will produce ultrasonic waves. The frequency of vibrations is

$$f = \frac{p}{2l} \cdot \sqrt{\frac{Y}{\rho}}$$

- Where, Y is Young's modulus, ρ is the density of the material, l is the length, and p=1,2,3 … For fundamental, first overtone, second overtone… respectively
- At resonance conditions,

$$f = \frac{1}{2\pi\sqrt{L_1 C_1}} = \frac{p}{2l} \cdot \sqrt{\frac{Y}{\rho}}$$

Merits:

- More efficient than a magnetostriction oscillator. Almost all the modern ultrasonic generators are of this type only.
- Frequency can be achieved up to 5×10^8 Hz
- O/p of this oscillator is very high.
- Not affected by temperature and humidity

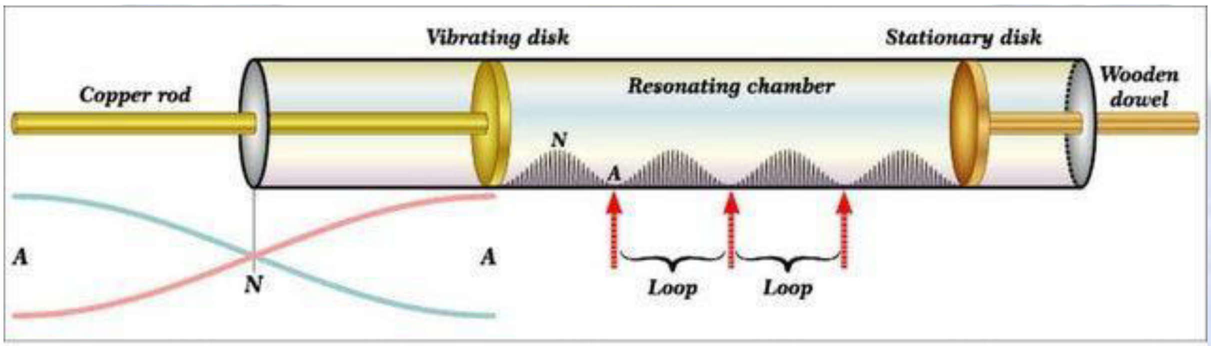
Demerits:

- Cost is high, and crystal's cutting and shaping are very complex.

Detection of ultrasonic waves:

1. Kundt's tube method
2. Thermal detector method.
3. Sensitive flame method.
4. Piezo-electric detector

1. Kundt's tube method



- Kundt's tube apparatus consists of a long circular glass tube of 1m in length and a diameter of 3 cm to 4cm.
- One end of the tube is fitted with an adjustable piston rod with a cork.
- The quartz crystal is placed between the two metal plates at the opening of the other end of the tube.
- Lycopodium powder is spread uniformly inside the tube.
- As in the case of sound waves, the Stationary supersonic Waves are produced in the air contained in a long glass tube, which is placed horizontally.
- The lycopodium powder gets collected in the form of heaps at the nodes and blown off at the antinodes.
- By measuring the average distance between the adjacent heaps, the wavelength and the Velocity can be calculated using the following relations
- The wavelength of the ultrasonic waves is given by $\lambda=2d$.
- where d is the distance between successive nodes.
- The ultrasonic velocity in the medium is $u = f\lambda$

$$= 2fd \text{ m/s}$$

Applications of Ultrasonic Waves

Science and Engineering:

- Used to detect flaws or cracks in metals
- Used to detect ships, submarines, icebergs, etc. in the ocean
- Used for soldering aluminium coil capacitors, aluminium wires, and plates without using any fluxes.
- used for cutting, drilling holes, and welding metals
- used as a catalytic agent and accelerates chemical reactions
- Also used to inspect any object without damage, which is called Non-Destructive Testing.

Medicine:

- Used to remove kidney stones and brain tumours without shedding any blood
- Used to remove broken teeth
- Used for sterilising milk and to kill bacteria
- Used to study the blood flow velocities in the blood vessels of our body
- Used as a diagnostic tool to detect tumours and for cancer treatment.
- **It is also used to find the depth of the sea through the SONAR technique.**

SONAR (Sound Navigation and Ranging):

- Because of the properties of ultrasonic waves, the Depth of the sea and any obstacles can be identified with the help of SONAR.
- It is based on the principle of echo sounding.
- The transmitter of SONAR will transmit the ultrasonic rays in various directions into the sea. These rays will get reflected when they are hit by any obstacles. This reflection will be picked up by the receiver.
- By measuring the distance and time of transmitted and reflected waves, the velocity of the ultrasonic wave is calculated.
- distance travelled by transmitted and reflected wave is AC + BC

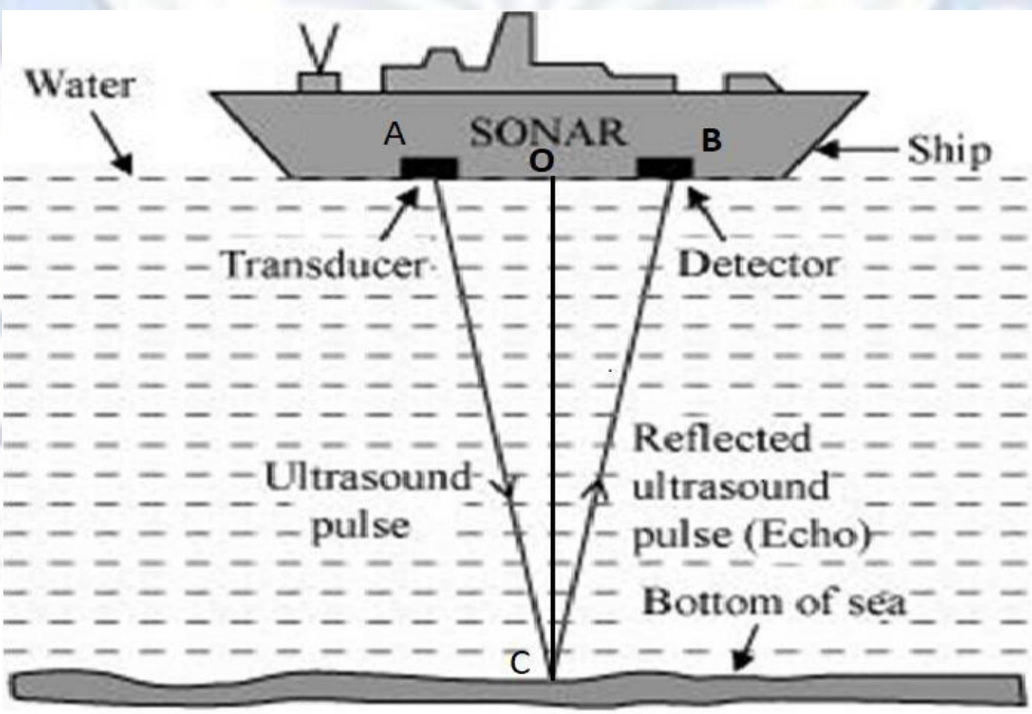
Velocity v =AC + BC/t,

so v = 2 CO/t,

So the depth of the sea is CO = v*t/2

- Using SONAR, the distance and direction of submarines, the depth of the sea, the depth of rocks in the sea, etc., can be determined.

- A fathometer or Echo meter is a device that is directly calibrated to determine the depth of the sea.



Non-destructive Testing

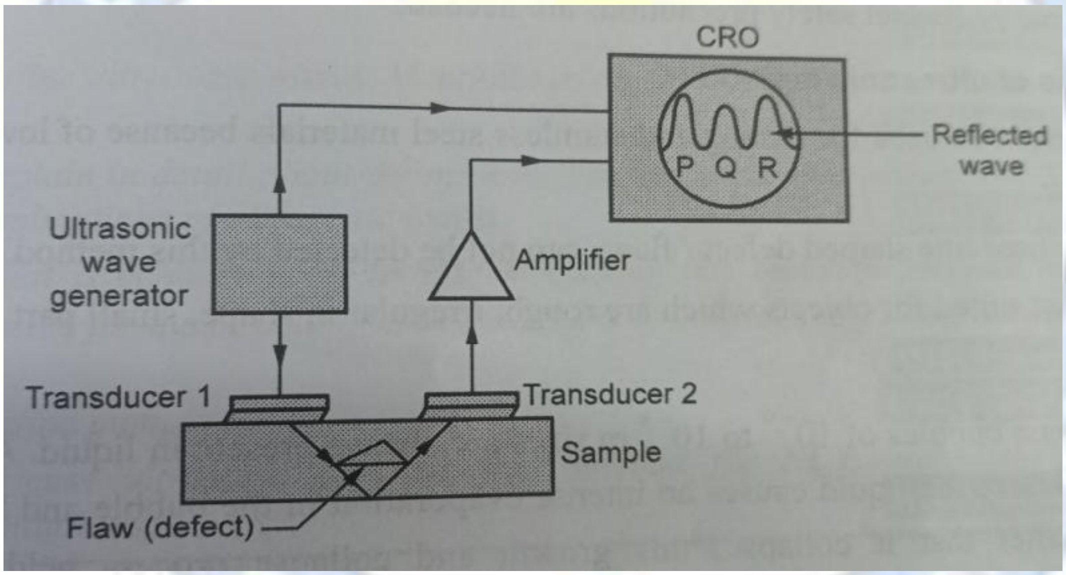
NDT is defined as the process of testing the material without causing any damage. Damage or reducing the service life of the component.

The general objectives of NDT:-

1. To increase serviceability
2. To improve productivity and to increase profits
3. To increase safety.

Flaw Detection Method.

- It is also based on each sounding. It helps to find the internal defect present in a sample.
- The ultrasonic wave generator transmits the wave, which gets reflected by the top and bottom surfaces of a sample and by a flaw.
- The receiving signal is seen/recorded by CRO.



- The ultrasonic wave generator produces the ultrasonic waves in the form of an electric pulse. This pulse gets converted into an optical signal by transducer 1.
- The optical signal propagated through the Sample. It gets reflected by a flaw if present in the sample and bottom of the sample.
- The reflected signal is again converted to an electrical pulse using transducer 2.
- The amplified signal /pulse is then observed on a CRO (cathode Ray Oscilloscope).
- The P, Q, and R three peaks are observed on the screen.

  P →Amplitude due to transmitted wave
  Q →Amplitude due to reflected wave from a flaw (defect)
  R →Amplitude due to the reflected wave from the bottom of the Sample.
- If the Q peak is present on CRO, then the flaw is present; otherwise, the Sample is flaw-free.
- Thus, using ultrasonic be can be detected from the Sample.
- **CRO- a device that displays and analyses electrical signals as a waveform.**

Advantages of Ultrasonic method

1. It can detect defects surface and inside
2. It is easy to operate.
3. The test is fast
4. It is highly accurate to detect the location, size and shape of the defect / Flaws.
5. The large thickness was also measured in this method
6. It gives an immediate result
7. The device required is portable
8. No special safety precautions are needed.

limitations

1. No permanent record of the flow can be obtained, and it can be observed only on the screen.
2. Only skilled and well-trained technicians can perform this testing
3. Parts that are rough in shape, very small or thin or not homogeneous are difficult to inspect
4. Discontinuities that are present in shallow layers immediately beneath the surface may not be detectable.